

EDUCATION

New York University Ph.D. in Psychology (Cognition and Perception)	New York, NY 2024–now
University of California, Irvine Ph.D. Program in Language Science Ph.D. Program in Logic and Philosophy of Science	Irvine, CA 2023–2024 2022–2023
University of Washington M.S. in Computational Linguistics	Seattle, WA 2020–2022
University of California, San Diego B.A. in Philosophy	La Jolla, CA 2016–2020

EXPERIENCE

Basis Research Institute Research Intern	New York, NY Summer 2026
Posh Technologies NLP Research Intern	Boston, MA Summer 2021

PUBLICATIONS

Journal articles and top-tier proceedings

- [1] N. Imel and N. Zaslavsky. Evolution and compression in LLMs: On the emergence of human-aligned categorization. In: *Proceedings of the International Conference on Learning Representations (ICLR) 2026*. Rio de Janeiro, Brazil, 2026. <https://openreview.net/forum?id=s7gSTR2AqA&>.
- [2] N. Imel, Q. Guo, and S. Steinert-Threlkeld. An efficient communication analysis of modal typology. *Open Mind* (2026). <https://doi.org/10.1162/OPML.a.313>.
- [3] S. Steinert-Threlkeld, N. Imel, and Q. Guo. A semantic universal for modality. *Semantics and Pragmatics* (2023). <https://doi.org/10.3765/sp.16.1>.

Peer reviewed proceedings and workshop papers

- [4] N. Imel and N. Zaslavsky. Culturally transmitted color categories in LLMs reflect a learning bias toward efficient compression. In: *CogInterp: Interpreting Cognition in Deep Learning Models (CogInterp @ NeurIPS)*. Best paper award. 2025. <https://arxiv.org/abs/2509.08093>.
- [5] N. Imel, J. Culbertson, S. Kirby, and N. Zaslavsky. Iterated language learning is shaped by a drive for optimizing lossy compression. In: *Proceedings of the 46th Annual Meeting of the Cognitive Science Society*. 2025. <https://escholarship.org/uc/item/63d7n4v0>.
- [6] M. Taliaferro, N. Imel, N. Zaslavsky, and E. Blanco-Elorrieta. Bilinguals exhibit semantic convergence while maintaining near-optimal efficiency. In: *Proceedings of the 46th Annual Meeting of the Cognitive Science Society*. 2025. <https://escholarship.org/uc/item/4128j529>.

- [7] N. Imel, C. Haberland, and S. Steinert-Threlkeld. The Unnatural Language Toolkit (ULTK). In: *Proceedings of the Society for Computation in Linguistics*. 2025. <https://doi.org/10.7275/scil.3144>.
- [8] N. Imel and N. Zaslavsky. Optimal compression in human concept learning. In: *Proceedings of the 46th Annual Meeting of the Cognitive Science Society*. 2024. <https://escholarship.org/uc/item/7pc1g61d>.
- [9] N. Imel and Z. Hafen. Citation-similarity relationships in astrophysics. In: *AI for Scientific Discovery: From Theory to Practice Workshop (AI4Science @ NeurIPS)*. 2023. <https://openreview.net/pdf?id=mISay7DPI>.
- [10] N. Imel, R. Futrell, M. Franke, and N. Zaslavsky. Noisy population dynamics lead to efficiently compressed semantic systems. In: *NeurIPS Workshop on Information-Theoretic Principles in Cognitive Systems Workshop (InfoCog @ NeurIPS)*. 2023. <https://openreview.net/pdf?id=9oEXdXmMnr>.
- [11] N. Imel. The evolution of efficient compression in signaling games. In: *Proceedings of the 45th Annual Meeting of the Cognitive Science Society*. 2023. <https://escholarship.org/uc/item/5dr5h4q0>.
- [12] N. Imel and S. Steinert-Threlkeld. Modals in natural language optimize the simplicity/informativeness trade-off. In: *Proceedings of Semantics and Linguistic Theory (SALT 32)*. 2022. <https://doi.org/10.3765/salt.v1i0.5346>.
- [13] Q. Guo, N. Imel, and S. Steinert-Threlkeld. A Database for Modal Semantic Typology. In: *Proceedings of the 4th Workshop on Research in Computational Linguistic Typology and Multilingual NLP*. Seattle, Washington: Association for Computational Linguistics, July 2022. <https://aclanthology.org/2022.sigtyp-1.6/>.

Works in progress

- [14] N. Imel, R. Futrell, M. Franke, and N. Zaslavsky. Agent-based imitation dynamics can yield efficiently compressed population-level vocabularies. (2026). <https://arxiv.org/abs/2603.15903>.
- [15] N. Imel and N. Zaslavsky. On convexity and efficiency in semantic systems. (2026). <https://arxiv.org/abs/2602.08238>.
- [16] N. Imel and Z. Hafen. Density, asymmetry and citation dynamics in scientific literature. (2025). <https://arxiv.org/abs/2506.23366>.
- [17] W. Uegaki, A. Mucha, N. Imel, and S. Steinert-Threlkeld. Deontic priority in the lexicalization of impossibility modals. (2023). <https://psyarxiv.com/h63y9/>.

TALKS AND PRESENTATIONS

Culturally transmitted color categories in LLMs reflect a learning bias toward efficient compression.

CogInterp @ NeurIPS 2025 (San Diego, CA)	12/07/25
EvoLang 2026 (Plovdiv, Bulgaria)	04/07/26
ICLR 2026 (Rio de Janeiro, Brazil)	04/25/26

Iterated learning is shaped by a drive for optimizing lossy compression

Cogsci 2025 (San Francisco, CA)	8/01/25
NYU C&P Miniconvention (New York, NY)	9/26/25
EvoLang 2026 (Plovdiv, Bulgaria)	04/07/26

Optimal compression in human concept learning

Cogsci 2024 (Rotterdam, NL)	7/24/24
PhilLab at Dartmouth College (virtual)	11/12/24

Noisy population dynamics lead to efficiently compressed semantic systems

University of Tübingen Linguistics Colloquium (virtual)

7/04/23

MIT Computational Psycholinguistics Lab (Cambridge, MA)

10/17/23

Evolang 2024 (Madison, WI)

5/19/24

Citation-similarity relationships in astrophysics literature

Santa Fe Institute Workshop on Intelligence and Representation (Cambridge, UK)

8/18/23

Modals in natural language optimize the simplicity/informativeness trade-off

Semantics and Linguistic Theory (Mexico City, CDMX)

6/08/22

Experiments in Linguistic Meaning (Philadelphia, PA)

5/18/22

AWARDS

Best paper award at CogInterp Workshop @ NeurIPS 2025

2025

Santa Fe Institute Complexity GAINs Summer Fellowship

2023

North American Summer School for Logic, Language and Information Student Fellowship (USC)

2023

Merit Fellowships (UC Irvine School of Social Sciences)

2022

Accepted to Summer School in Logic and Formal Epistemology (CMU)

2021

Best paper “Desire Semantics”, selected for UC San Diego undergraduate philosophy journal *Intuitions*

2020

TEACHING

Grader

- ANOVA (NYU) – w. Ajua Duker

Fall 2025

Teaching Assistant

- Lab in Cognition and Perception (NYU) – w. Noga Zaslavsky

Spring 2025

- Introduction to Syntax (UCI) – w. Galia Bar-Sever

Winter, Spring 2024

- Introduction to Linguistics (UCI) w. Ben Mis

Fall 2023

- Basic Economics I (UCI) w. Jiawei Chen

Spring 2023

- Introduction to Symbolic Logic – w. Toby Meadows (UCI)

Winter 2023

SERVICE

Organizing

- Co-organizer for InfoCog Workshop @ CogSci

2025

- Organizing Committee for Society for Computation in Linguistics Conference

2023

- Co-organizer for InfoCog Workshop @ Neurips

2023

- Department Colloquium Committee (UCI Language Science)

2023

Reviewing

- Journal reviewing: JML, Mind & Language

2024-2025

- Conferences/Workshops: CogSci, InfoCog Workshop @ NeurIPS, EvoLang, California Annual Meeting of Psycholinguistics

2023-2025

SKILLS

- **Programming Languages:** Python, Java, C++, MATLAB
- **Libraries & Frameworks:** NumPy, PyTorch, pandas, plotnine, matplotlib, huggingface, NLTK, Hydra
- **Cloud & Computing:** high-performance computing with Condor & Slurm, Google Cloud Platform, Lambda AI
- **DevOps & Tools:** GitHub Workflows/Actions, containerization (Singularity)